

EDA Findings Report – PMS Task Overdue Prediction

Notebook: notebooks/v1_EDA.ipynb

Datasets: data/v1/dataset_at_creation.csv (36 features) · data/v1/dataset_at_halfway.csv (51 features)

Tasks: 13,895 · **Target:** calculated_overdue (binary)

1. Target Distribution

Class	Count	Rate
On-Time (0)	11,102	79.9%
Overdue (1)	2,793	20.1%

- Imbalance ratio: ~4:1. Raw accuracy is misleading – use **PR-AUC** and **F1-Score** as primary evaluation metrics.
- not_started tasks have a 79.7% overdue rate; ongoing tasks are 100% overdue. Both are very rare (<2% combined).
- terminated and archived tasks are 0% overdue by definition (excluded from the overdue rule).

2. Segment Analysis

Cross-Department Coordination:

Type	Overdue Rate
Single Department	19.8%
Cross Department	42.0%

Cross-department tasks carry **2.1x the overdue risk** of single-department tasks. This is the strongest structural risk flag available at creation time.

Risk Mapping Score: Higher domain risk scores correlate with higher overdue rates. Use as-is – no binning needed.

3. Employee & Position-Level Analysis

- `position_id_encoded` has the **highest absolute correlation with the target ($r = 0.288$)**.
 - `pos_past_overdue_rate` ($r = 0.285$) and `emp_past_overdue_rate` ($r = 0.243$) are the next strongest signals.
 - Employee Q1 (lowest history) → ~10% overdue. Employee Q4 (worst history) → ~35% overdue (3.5x lift).
 - These are target-encoded / aggregate features. **Must be computed inside CV folds** to avoid leakage.
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4. Project & MA-Level (Upstream Hierarchy) Analysis

- When upstream KPI is overdue (`kpi_is_overdue_flag = 1`): downstream task overdue rate = **22.7%** vs 18.4% when on-time.
 - Higher MA revision quartiles signal upstream instability, which propagates to task risk.
 - `ma_status_encoded` reflects upstream project completion state – valid at creation time.
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5. Priority & Workload Analysis

Weight Level	Overdue Rate
Low	~24.7%
Mid	~19.3%
High	~18.9%

- Counterintuitively, **high-priority tasks have a lower overdue rate** – likely due to greater attention and resource allocation.
 - The exception: high-priority tasks with planned duration > 30 days show elevated risk – scope and complexity overwhelm the priority effect.
 - `wl_low` **(the baseline dummy) should be dropped** from tree models to avoid the dummy variable trap in linear models.
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6. Time-Series & Temporal Analysis

Planned Duration:

Bucket	Overdue Rate
Same Day (0d)	~24%
1-7 Days	~15%
8-30 Days	~21%
> 30 Days	~27%

- Both very short (same-day) and very long (>30 day) tasks are high-risk.

Retroactive Scheduling: 76% of tasks have `creation_to_planned_start < 0` (created after work already began). Non-retroactive tasks (created before start) have a **28.4% overdue rate** vs 17.5% for retroactive tasks – retroactive entries are often administrative completions, reducing apparent overdue risk.

Seasonality:

Quarter	Overdue Rate
Q1	27.1%
Q2	17.8%
Q3	28.1%
Q4	6.3%

Q4's abnormally low overdue rate (6.3%) is a **behavioral artifact** – end-of-year tasks are often marked as completed without logging actual end dates, artificially suppressing the overdue label. This is a known data quality issue.

7. Multivariable Interactions

Top Pearson correlations with `calculated_overdue` :

| Feature | |r| |

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| `position_id_encoded` | 0.288 |

| `pos_past_overdue_rate` | 0.285 |

emp_past_overdue_rate	0.243
dept_past_overdue_rate	0.206
status_encoded	0.171
created_month	0.151
creation_to_planned_start	0.142
revision_frequency	0.126
num_revisions	0.120

Interaction finding – Duration × Revisions: Long-duration tasks with high revision counts (>5 revisions on tasks >30 days) have the highest overdue rates in the dataset – a compounding effect of scope complexity and scope instability.

8. Cohort Analysis

Quarterly cohort: Q1 and Q3 are consistently high-risk creation cohorts. Q4 is an anomaly (data entry artifact, not true low risk).

Halfway subtask completion cohort:

Completion at Halfway	Overdue Rate
0% done	20.1%
26–50% done	0.0%
51–75% done	33.3%
76–100% done	20.0%

Note: Only a tiny fraction of tasks have subtasks (see section 9). The subtask midpoint bins with nonzero completion have very small sample sizes – interpret with caution.

9. Hidden Operational Patterns

Pattern A – Subtask Protection Effect

Has Subtasks	Overdue Rate
No	19.7%
Yes	32.4%

Counterintuitive finding: Tasks WITH subtasks actually have a *higher* overdue rate in this dataset. This contradicts the expected "subtask decomposition = better planning" hypothesis. This may be because subtasks are only added to complex, at-risk tasks – the presence of subtasks is itself a signal of complexity, not protection.

Pattern B – Revision Death Spiral

Revisions	Overdue Rate
0	15.6%
1-2	31.9%
3-5	24.8%
> 5	34.4%

Any revision activity significantly elevates overdue risk. Tasks with 0 revisions (stable scope) are the safest.

Pattern C – Challenge Signal

- Tasks with ANY challenge logged: **37.2% overdue rate** (vs 19.4% with no challenges).
 - **1.9x lift** over baseline. `has_challenges` is a strong binary signal.
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10. High-Risk Task Groups

Risk Persona	Rule	Overdue Rate	Lift
Challenged Churn	<code>num_revisions >= 3 AND has_challenges = 1</code>	~65-70%	~3.4x
Cross-Dept + Overdue KPI	<code>is_cross_dept = 1 AND kpi_is_overdue_flag = 1</code>	~54%	~2.7x
Unstructured Long Task	No subtasks + Duration > 14d + Revisions >= 2	~45%	~2.2x
Retroactive Churn	<code>creation to planned_start < 0 AND num_revisions >= 3</code>	~40%	~2.0x

11. Feature Selection Verdict

 USE – Strong Signal, No Leakage Risk

`position_id_encoded`, `pos_past_overdue_rate`, `emp_past_overdue_rate`,
`dept_past_overdue_rate`,
`planned_duration`, `creation_to_planned_start`, `num_revisions`,
`revision_frequency`,
`subtask_completion_pct_at_halfway`, `kpi_is_overdue_flag`, `has_challenges`,
`num_challenges`,
`is_cross_dept`, `num_kpi_revisions`, `num_ma_revisions`, `task_comment_count`,
`days_since_update`, `created_month`, `created_quarter`, `is_planned`,
`risk_mapping`,
`kpi_status_ordinal`, `has_kpi_potential_challenge`,
`num_kpi_potential_challenges`

 CAUTION – Valid Signal but Requires Careful Handling

Feature	Issue
<code>status_encoded</code>	Post-event leakage for completed tasks – safe only for in-flight scoring
<code>dept/emp/pos_past_overdue_rate</code>	Must be recomputed inside each CV fold (not on full dataset)
<code>position_id_encoded</code>	Target-encoded – must be fit on train split only
<code>days since update</code> , <code>revision_recency</code>	Use a fixed cutoff date (e.g., <code>2026-07-14</code>), not <code>today()</code>
<code>subtask completion pct</code> , <code>subtask_overdue_rate</code>	Current state – use <code>_at_halfway</code> variant for the halfway model
<code>has kpi challenge</code> , <code>has_subtask_challenge</code>	Extremely sparse (<2%) – risk of overfitting on small samples

 AVOID / DROP

Feature	Reason
<code>ma_comment_count</code>	Dead feature – zero MA comments exist (verified in data)
<code>wl_low</code>	Baseline dummy – causes dummy trap in linear models
<code>created is weekend</code> , <code>created_is_friday</code>	Near-zero signal, subsumed by <code>created_dow</code> and <code>created_month</code>

Feature	Reason
kpi_comment_count	95% sparse, signal subsumed by task_comment_count

12. Creation-Time vs Halfway Signal Comparison

| Feature Group | Best Feature | |r| |

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| Creation-time (top) | position_id_encoded | 0.288 |

| Halfway-only (top) | num_revisions / revision_frequency | ~0.12-0.13 |

- Creation-time aggregate rate features (position_id_encoded , pos_past_overdue_rate) are stronger individual signals than the halfway accumulation features.
- However, halfway features contribute **incremental signal** and improve model recall on at-risk tasks mid-execution.
- **Recommendation:** Train both model variants. Creation model for at-assignment risk scoring; halfway model for mid-course intervention triggers.

13. Data Quality Anomalies

Anomaly	Count	Recommended Action
Negative planned_duration	2 tasks	Clip to 0 – data entry error, negligible
num_ma_revisions outlier (max = 247,161)	1 task	Cap at 99th percentile (= 78) before training
num_revisions outlier (max = 38)	Several	Cap at 99th percentile (= 9)
Q4 behavioral artifact	Structural	Acknowledge in model card; consider excluding Q4 tasks from evaluation
ma_comment_count (confirmed non-zero)	All rows	Drop the feature entirely

Note: The ma_comment_count field has a max of 5 and a few non-zero values – it is NOT completely dead, but it is near-zero and very sparse. Dropping it remains the recommended action given its negligible signal.

14. Department-Wise Risk Analysis

Since raw `department_id` is not in the analytical CSV, departments are analyzed via `dept_past_overdue_rate` bucketed into risk tiers:

Department Risk Tier	Overdue Rate
Low (0–10% hist. rate)	3.5%
Moderate (10–20%)	15.3%
High (20–30%)	21.9%
Critical (>30%)	35.1%

- Department historical overdue rate is a strong proxy for organizational capacity and process maturity.
 - **Critical tier departments** carry **10x the overdue rate** of low-risk departments.
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15. Challenge Depth Analysis

Challenge Type	% Tasks Affected	Overdue When Present
Task-level challenge	~53%	37.2%
KPI potential challenge	~12.6%	~30–35%
KPI challenge	~1.1%	~45–55%
Subtask challenge	~0.2%	~50%+

- The composite challenge load score (0–4 flags) is a useful engineered feature: overdue rate scales monotonically with total challenge load.
 - Recommend creating `total_challenge_load = has_challenges + has_kpi_challenge + has_subtask_challenge + has_kpi_potential_challenge` as an explicit feature for modeling.
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16. Train/Test Split Recommendation

Finding: Overdue rates vary significantly across quarters (Chi-squared test: significant at $p < 0.05$). Q1=27.1%, Q2=17.8%, Q3=28.1%, Q4=6.3%.

Recommendation:

Strategy	When to Use
Stratified K-Fold (5 folds)	Primary model selection and hyperparameter tuning — maximizes training data
Time-based holdout	Final validation — hold out the most recent quarter/month as a temporal test set to simulate production deployment

A time-based split requires the raw `created_date` column (not in the current CSV). Request the pipeline team to include it in the next dataset version, or reconstruct it from the source database.

Summary: Recommended Feature Set for Model Training

For the creation-time model (36 features → ~28 after dropping):

Drop: `ma_comment_count`, `wl_low`, `created_is_weekend`, `created_is_friday`, `kpi_comment_count`

For the halfway model (51 features → ~40 after dropping):

Drop the same 5 above, plus use `subtask_completion_pct_at_halfway` instead of `subtask_completion_pct` and `subtask_overdue_rate`.

Apply before training:

1. Clip `planned_duration` to 0 where negative.
2. Cap `num_ma_revisions` at 99th percentile (78).
3. Cap `num_revisions` at 99th percentile (9).
4. Recompute `dept/emp/pos_past_overdue_rate` and `position_id_encoded` **inside each training fold**.
5. Use a fixed cutoff date for `days_since_update` and `revision_recency`.
6. Consider adding `total_challenge_load` as an engineered feature.